Lotus Leaf Surface Sprint

Build a surface that sheds water and dirt the way a lotus leaf does.

Win condition: Best clean runoff and least residue after one redesign.

THE SCIENCE YOU ARE USING

Hydrophobic micro-texture. A lotus leaf is covered in tiny bumps coated in wax. Water cannot settle into the texture, so droplets bead up into near-spheres and roll off, carrying dirt with them. The effect comes from roughness plus a water-repelling coating, not from being smooth.

Roughness plus coating. You texture and coat safe surfaces, then race droplets down a standard ramp. The best surface sheds water fast and leaves the least residue. Copying the leaf on purpose is biomimicry in materials engineering.

HOW TO BUILD AND RUN IT

01 **Compare bare surfaces.** Drop water on smooth and rough surfaces. Watch which makes the droplet bead up.

02 **Texture and coat.** Add micro-texture and a water-repelling layer to your test surface.

03 **Add a dirt challenge.** Sprinkle pepper or cocoa as fake dirt. A good surface carries it off with the droplet.

04 **Race down the ramp.** Run droplets down the standard ramp. Time the runoff and check the residue left behind.

05 **Redesign for less residue.** Adjust texture or coating to shed cleaner, then re-run.

Safety: Work over a tray and wipe spills right away so the floor stays dry. Use only the washable dirt proxy (pepper or cocoa) and classroom-safe wax (wax paper, or a candle rubbed on cold and never lit). Brush off loose sandpaper grit. No goggles are needed for dropping water onto a tilted surface. Allergy: Cocoa allergy substitute: pepper or washable soil.

MATERIALS

Wax paper	shared
Cardstock	shared
Sandpaper strips	shared
Masking tape	shared
Pipettes	per team (3 mL graduated, reused)
Water	shared (tap)
Pepper or cocoa	residue (washable dirt proxy; source locally)
Candle or paraffin wax	shared (rubbed on cold to coat a surface; source locally)
Ramps	per team (on-hand rigid board at a fixed tilt)

YOUR PLAN AND DATA

Clean, complete data is worth as much as a fast result.

Prediction (what will work best, and why): ______________________________

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How we built our surface (texture + coating we used): ______________________________

The ONE thing we changed for the redesign (texture, amount of wax, tilt, or material): ______________________________

Runoff time: baseline ________ **after redesign** ________ **Residue left: baseline** ________ **after redesign** ________

DEFEND YOUR DESIGN

What evidence made you change your design? ______________________________

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If you ran it again, what is the ONE thing you would change? ______________________________

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HOW YOU SCORE (100 POINTS)

SCORING CRITERION	POINTS
Fast clean runoff	35
Least residue	25
Surface design explanation	20
Redesign gain	10
Craftsmanship	10
Total	**100**